

User Manual

WorkWithHelmtOnly

Construction Safety Monitoring System

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1. Introduction

1.1 System Overview

WorkWithHelmtOnly is an advanced construction safety monitoring system that uses computer vision technology to automatically detect and track workers' compliance with safety equipment requirements. The system processes video feeds (live or recorded) to identify workers and determine whether they are wearing required safety equipment, specifically focusing on helmets and safety vests.

The system combines state-of-the-art object detection using YOLO (You Only Look Once) with real-time tracking using SORT (Simple Online and Realtime Tracking) to maintain consistent monitoring of individuals even as they move throughout a construction site. When safety violations are detected, the system logs the events, generates visual alerts, and can send email notifications to safety supervisors.

1.2 Purpose

The primary purpose of WorkWithHelmtOnly is to enhance workplace safety by:

1. Automating the monitoring of safety equipment compliance
2. Providing real-time alerts for safety violations
3. Generating analytical reports to identify patterns and trends in compliance
4. Creating a documented record of safety practices for regulatory purposes
5. Reducing the workload on safety managers through automated monitoring

By implementing this system, construction companies can significantly reduce the risk of workplace injuries, improve compliance with safety regulations, and create a data-driven approach to safety management.

1.3 Main Features

WorkWithHelmtOnly offers the following key features:

- **Real-time Detection:** Process live webcam feeds to detect workers and safety equipment
- **Video File Analysis:** Upload and analyze pre-recorded video files
- **Worker Tracking:** Maintain consistent tracking of individuals across video frames
- **Compliance Monitoring:** Automatically detect whether workers are wearing required safety equipment
- **Violation Alerts:** Generate real-time visual alerts and email notifications for safety violations
- **Comprehensive Analytics:** Visualize compliance data through charts, graphs, and statistics
- **Customizable Dashboard:** Intuitive interface with configurable settings
- **Data Export:** Export compliance data for reporting and integration with other systems

2. System Requirements

2.1 Hardware Requirements

Minimum Requirements

- **Processor:** Intel Core i5 (8th generation) or AMD Ryzen 5 or equivalent
- **RAM:** 8GB
- **Graphics:** Integrated GPU (Intel UHD or AMD equivalent)
- **Storage:** 10GB free disk space
- **Camera:** 720p webcam or IP camera (for live monitoring)
- **Display:** 1366x768 resolution

Recommended Requirements

- **Processor:** Intel Core i7 (10th generation) or AMD Ryzen 7 or equivalent
- **RAM:** 16GB
- **Graphics:** NVIDIA GTX 1650 or better (with CUDA support)
- **Storage:** 20GB SSD storage
- **Camera:** 1080p webcam or IP camera with good low-light performance
- **Display:** 1920x1080 resolution or higher

2.2 Software Requirements

- **Operating System:**
 - Windows 10/11 (64-bit)
 - Ubuntu 20.04 LTS or later
 - macOS 11 (Big Sur) or later
- **Dependencies:**
 - Python 3.8 or later
 - CUDA 11.4+ (for NVIDIA GPU acceleration, optional)
 - Git (for installation and updates)
- **Web Browser** (for Streamlit interface):
 - Google Chrome (recommended)
 - Mozilla Firefox
 - Microsoft Edge (Chromium-based)
 - Safari

2.3 Network Requirements

- **Bandwidth:** Minimum 1 Mbps upload/download for webcam streaming
- **Connectivity:** Stable internet connection for email notifications
- **Ports:** 8501 (for Streamlit web interface)
- **Email Server Access:** SMTP access for sending notifications

3. Installation

3.1 Prerequisites

Before installing WorkWithHelmtOnly, ensure the following prerequisites are met:

1. Python 3.8 or later is installed on your system
2. Git is installed for cloning the repository
3. pip (Python package installer) is up to date
4. Virtual environment tool (venv or conda) is installed (recommended)
5. CUDA toolkit is installed (for GPU acceleration with NVIDIA graphics cards)

3.2 Installation Steps

Follow these steps to install the WorkWithHelmtOnly system:

Step 1: Clone the Repository

Open a terminal or command prompt and run:

```
git clone https://github.com/AbdullahAboElHija/FinalProject.git
cd FinalProject
```

Step 2: Create a Virtual Environment (Recommended)

For Windows:

```
python -m venv venv
venv\Scripts\activate
```

For macOS/Linux:

```
python3 -m venv venv
source venv/bin/activate
```

Step 3: Install Required Dependencies

Install all required packages using the requirements.txt file:

```
pip install -r requirements.txt
```

Step 4: Download the YOLO Model

The system will automatically download the required YOLO model weights on the first run. Alternatively, you can manually download the model file:

```
mkdir -p models
```

```
cd models

wget
https://github.com/AbdullahAboElHija/FinalProject/releases/download/v1.0/ppe.pt

cd ..
```

Step 5: Verify Installation

Run the following command to verify that all dependencies are correctly installed:

```
python -c "import ultralytics, numpy, matplotlib, seaborn, streamlit, cv2,
filterpy; print('All dependencies successfully installed!')"
```

3.3 Troubleshooting Installation

If you encounter issues during installation, try the following solutions:

- **Dependency Errors:**
 - Make sure pip is up to date: `pip install --upgrade pip`
 - Try installing dependencies one by one to identify problematic packages
 - Check compatibility between Python version and package versions
- **CUDA Issues:**
 - Verify CUDA installation with: `nvcc --version`
 - Ensure PyTorch is installed with CUDA support: `python -c "import torch; print(torch.cuda.is_available())"`
 - Check CUDA version compatibility with installed PyTorch version
- **Model Download Issues:**
 - Check internet connection
 - Try manual download from the GitHub repository
 - Verify disk space availability
- **Permission Issues:**
 - Run terminal/command prompt as administrator (Windows) or use `sudo` (Linux/macOS)
 - Check write permissions in the installation directory

4. System Configuration

4.1 Initial Setup

When running the application for the first time, several configuration steps are recommended:

1. **Create profile:** Navigate to the "About Me" tab and set your username and profile picture
2. **Configure email:** Set up email notification settings in the "Set Receiver Email" tab
3. **Test webcam:** Verify webcam functionality in the "Dashboard" tab before deployment
4. **Adjust FPS:** Configure the frame processing rate based on your hardware capabilities

4.2 Email Notification Setup

To enable email notifications for safety violations:

1. Navigate to the "Set Receiver Email" tab in the dashboard
2. Enter the email address where violation notifications should be sent
3. Click "Save" to update the notification settings

Note: The system uses a default sender email address (workwithhelmtonly@gmail.com). To use a custom sender address, modify the `sender_email` and `sender_password` variables in the source code.

4.3 Camera Configuration

Webcam Setup

1. Ensure your webcam is properly connected and recognized by your system
2. Position the webcam to capture a clear view of the monitored area
3. Adjust lighting to ensure good visibility of workers and safety equipment
4. Navigate to the Dashboard tab > Webcam sub-tab
5. Use the FPS slider to adjust frame rate according to your hardware capabilities

Video File Requirements

For analyzing pre-recorded videos:

- **Supported formats:** MP4, AVI, MOV
- **Recommended resolution:** 720p (1280x720) or higher
- **Maximum file size:** 500MB
- **Maximum duration:** 30 minutes
- **Frame rate:** 24-30 FPS recommended

4.4 Detection Parameters

The system uses predefined detection parameters that work well in most environments. Advanced users can modify these parameters in the source code:

- **Confidence threshold:** Default 0.4 (increase for fewer false positives, decrease for higher sensitivity)
- **IOU threshold:** Default 0.45 (adjust for overlapping detection handling)
- **Maximum detections:** Default 50 objects per frame
- **Tracking parameters:** Maximum age of 1 frame, minimum hits of 3 frames for confirmed tracks

5. Using the System

5.1 Starting the Application

To start the WorkWithHelmetOnly application:

1. Open a terminal or command prompt
2. Navigate to the project directory
3. Activate the virtual environment (if used)
4. Run the application with:
`streamlit run webapp.py`
5. The application will start, and your default web browser should open automatically to the application interface (typically at <http://localhost:8501>)

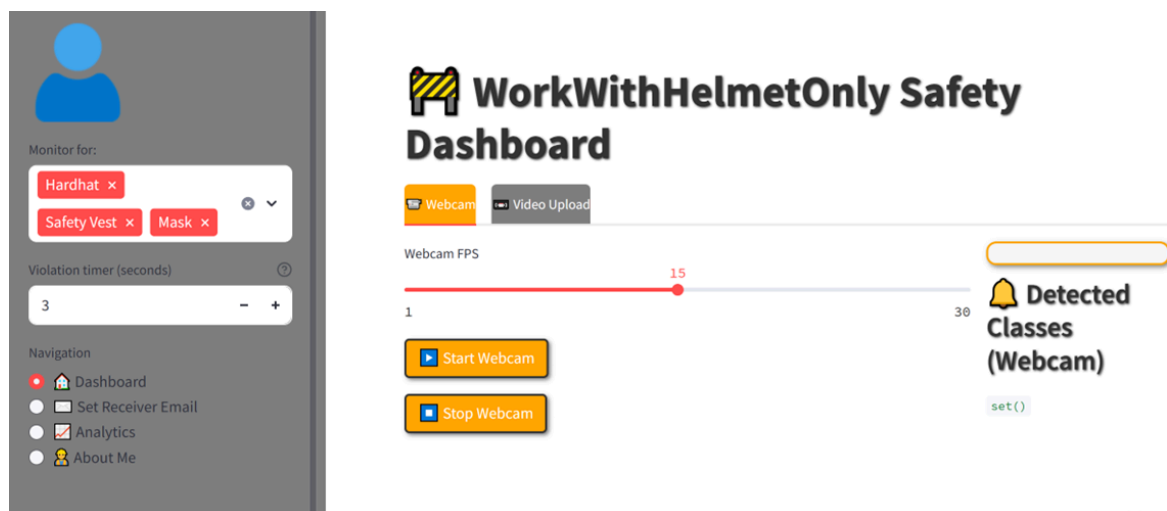
If the browser doesn't open automatically, copy the URL displayed in the terminal and paste it into your browser's address bar.

5.2 Dashboard Overview

The WorkWithHelmetOnly dashboard consists of four main tabs:

1. **Dashboard Tab:** The main monitoring interface for real-time webcam and video file analysis
2. **Set Receiver Email Tab:** Configuration for email notifications
3. **Analytics Tab:** Data visualization and compliance reporting
4. **About Me Tab:** User profile management

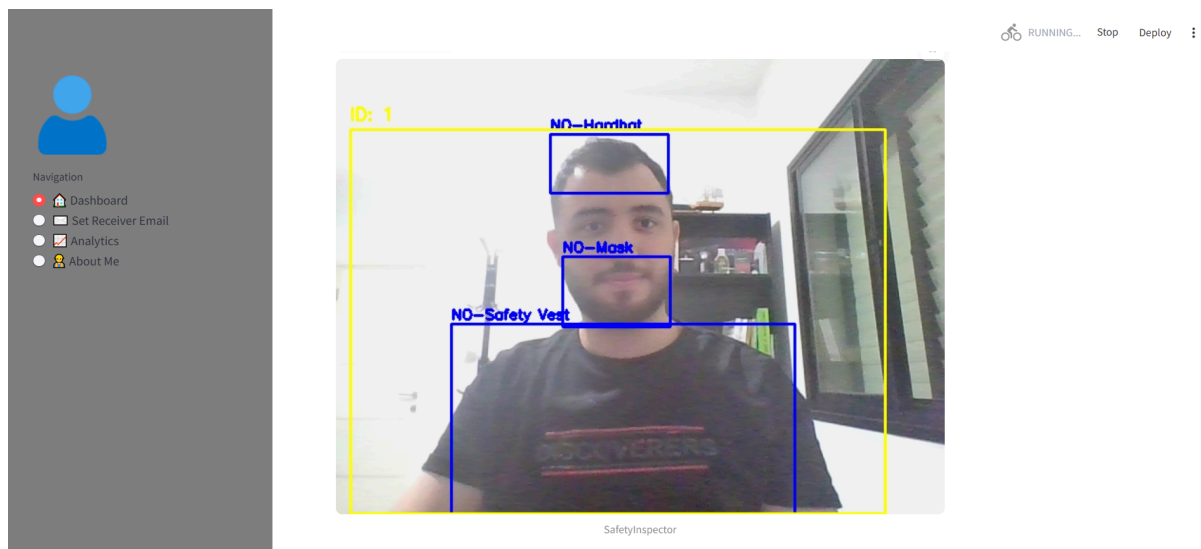
Navigate between tabs by clicking on the desired tab in the left sidebar. The sidebar also displays your profile image and username.



5.3 Real-time Monitoring

To monitor safety compliance in real-time using a webcam:

1. Navigate to the **Dashboard Tab** and select the **Webcam** sub-tab
2. Adjust the **FPS** slider to optimize performance for your hardware
3. Click **Start Webcam** to begin monitoring
4. The system will display the live feed with annotations:
 - Green boxes: Safety equipment detected
 - Blue boxes: Missing safety equipment
 - Yellow boxes: Tracked workers with ID numbers
5. Safety violations are automatically logged and may trigger email notifications
6. Click **Stop Webcam** to end the monitoring session and save data



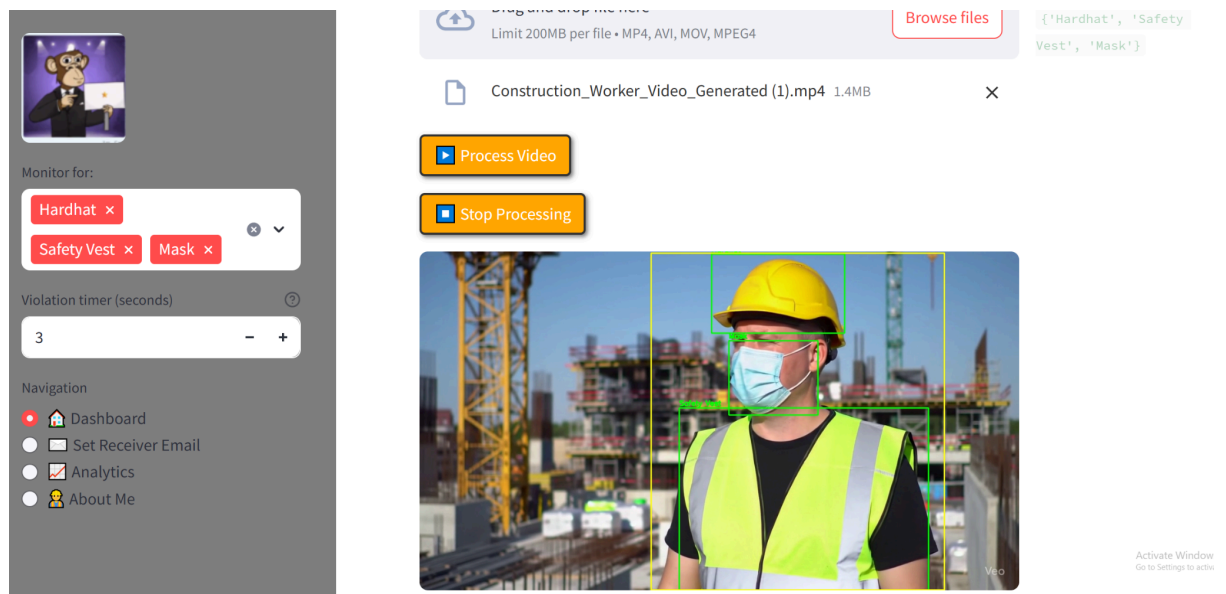
Tips for effective monitoring:

- Ensure good lighting in the monitored area
- Position the camera to minimize occlusions
- Keep the camera stable for better tracking
- Monitor the FPS counter in the interface to ensure smooth performance

5.4 Video Upload and Analysis

To analyze a pre-recorded video file:

1. Navigate to the **Dashboard Tab** and select the **Video Upload** sub-tab
2. Adjust the **FPS** slider to control processing speed
3. Click **Upload a video** and select a video file from your computer
4. After the video is uploaded, click **Process Video** to begin analysis
5. The system will display the video playback with annotations similar to the webcam view
6. You can click **Stop Processing** at any time to end the analysis early
7. After processing completes or is stopped, the system saves the analysis data



Supported video formats:

- MP4 (.mp4)
- AVI (.avi)
- MOV (.mov)

Notes:

- Large video files may take significant time to upload and process
- Processing time depends on video length, resolution, and your hardware capabilities
- For optimal performance, use videos with resolution 1280x720 or higher

5.5 Viewing Analytics

The Analytics tab provides insights into safety compliance data:

1. Navigate to the **Analytics Tab**
2. Select between **Webcam Data** and **Video Data** sub-tabs
3. View visualizations showing:
 - Violations by person ID (bar charts)
 - Safety check summary (pass/fail counts)
 - Raw data tables with detailed compliance information

Available visualizations:

- **Violations by Person ID:** Bar chart showing frequency of different violations by worker
- **Safety Check Summary:** Bar chart showing overall pass/fail rates
- **Raw Data Table:** Detailed listing of all monitoring events with timestamps

Filtering options:

- Filter by date range
- Filter by person ID
- Filter by violation type

Webcam Data - Safety Check Summary

	Person ID	Safety Check
0	1	Fail

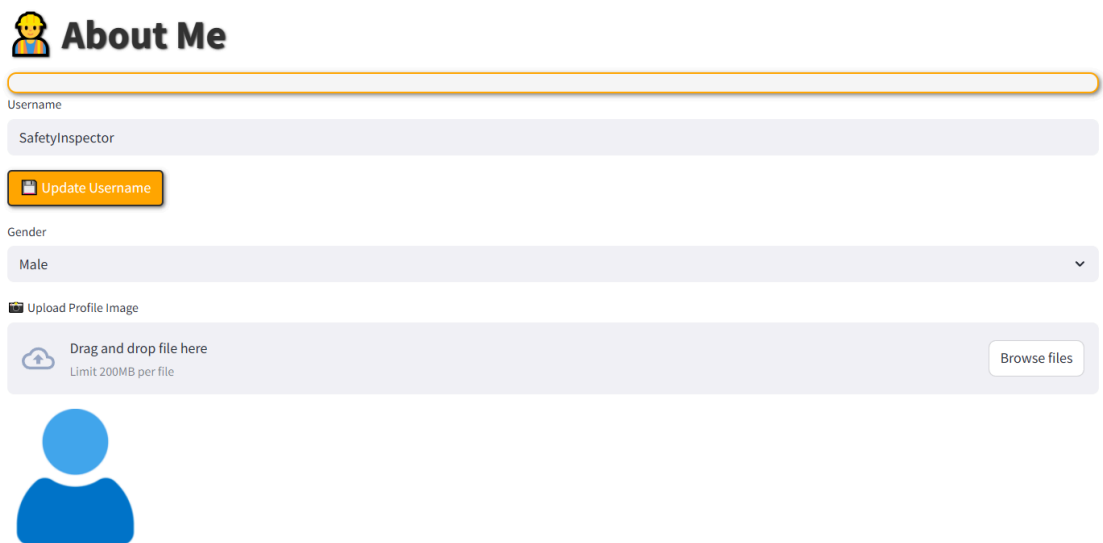
Webcam Data - Raw Data

	Timestamp	Person ID	Equipment Worn	Equipment Not Worn
0	2025-05-21 19:04:11	1	None	NO-Hardhat NO-Safety Vest NO-Mask

5.6 Managing User Profiles

To update your user profile:

1. Navigate to the **About Me Tab**
2. Update your **Username** in the provided field
3. Click **Update Username** to save changes
4. To update your profile picture, click **Browse** under "Upload Profile Image"
5. Select an image file from your computer
6. The new profile image will be displayed in the sidebar



The image shows a user profile update interface titled "About Me" with a yellow hard hat icon. It contains a text input for "Username" with the value "SafetyInspector" and an orange "Update Username" button. Below is a "Gender" dropdown menu set to "Male". The "Upload Profile Image" section features a drag-and-drop area with a cloud icon, the text "Drag and drop file here", and "Limit 200MB per file". A "Browse files" button is on the right. At the bottom is a blue circular profile picture placeholder.

About Me

Username

SafetyInspector

 Update Username

Gender

Male

 Upload Profile Image

 Drag and drop file here
Limit 200MB per file

 Browse files



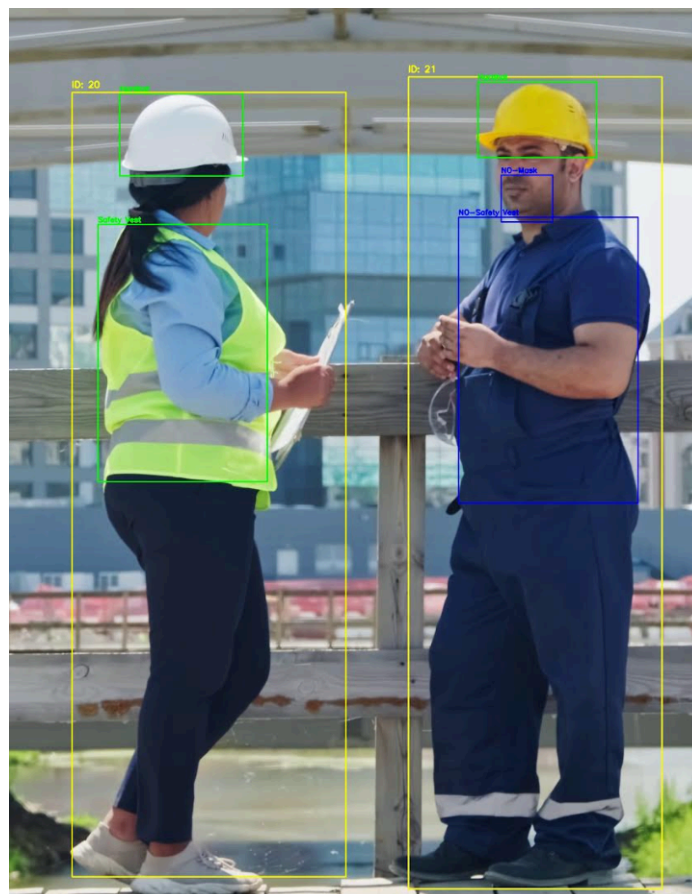
6. Interpreting Results

6.1 Detection Visualizations

The system uses color-coded visualizations to indicate safety status:

- **Green Bounding Boxes:** Detected safety equipment (Helmet, Safety Vest)
- **Blue Bounding Boxes:** Detected missing safety equipment (NO-Helmet, NO-Safety Vest)
- **Yellow Bounding Boxes:** Tracked workers with ID numbers
- **Text Labels:** Show the class of detected objects with confidence percentages

When a worker is non-compliant, the specific violation is indicated in the visualization.



6.2 Compliance Reports

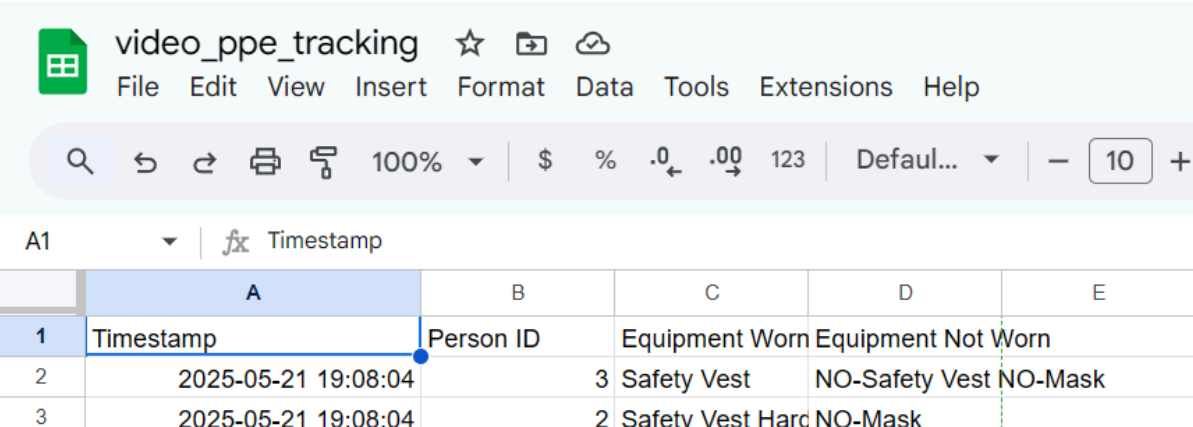
Compliance data is recorded in CSV files:

- **webcam_ppe_tracking.csv**: For webcam monitoring sessions
- **video_ppe_tracking.csv**: For video file analysis

Each record includes:

- **Timestamp**: Date and time of the detection
- **Person ID**: Unique identifier for the tracked worker
- **Equipment Worn**: List of detected safety equipment
- **Equipment Not Worn**: List of missing safety equipment

These CSV files can be opened in spreadsheet applications like Microsoft Excel or Google Sheets for further analysis or report generation.



	A	B	C	D	E
1	Timestamp	Person ID	Equipment Worn	Equipment Not Worn	
2	2025-05-21 19:08:04	3	Safety Vest	NO-Safety Vest	NO-Mask
3	2025-05-21 19:08:04	2	Safety Vest Harc	NO-Mask	

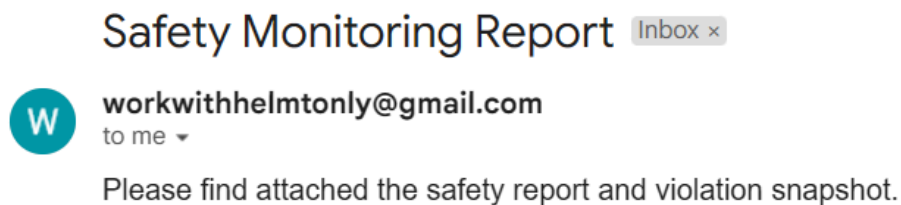
6.3 Alert Notifications

When a safety violation is detected, the system can send email notifications to the configured recipient. These notifications include:

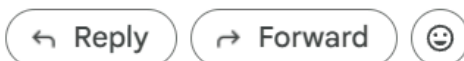
- **Subject:** Safety Monitoring Report
- **Body:** Brief description of the violation
- **Attachments:**
 - CSV file with violation details
 - JPEG image showing the violation

Email notifications are sent when:

- A monitoring session ends with detected violations
- (Optional) Immediately when violations are detected during monitoring



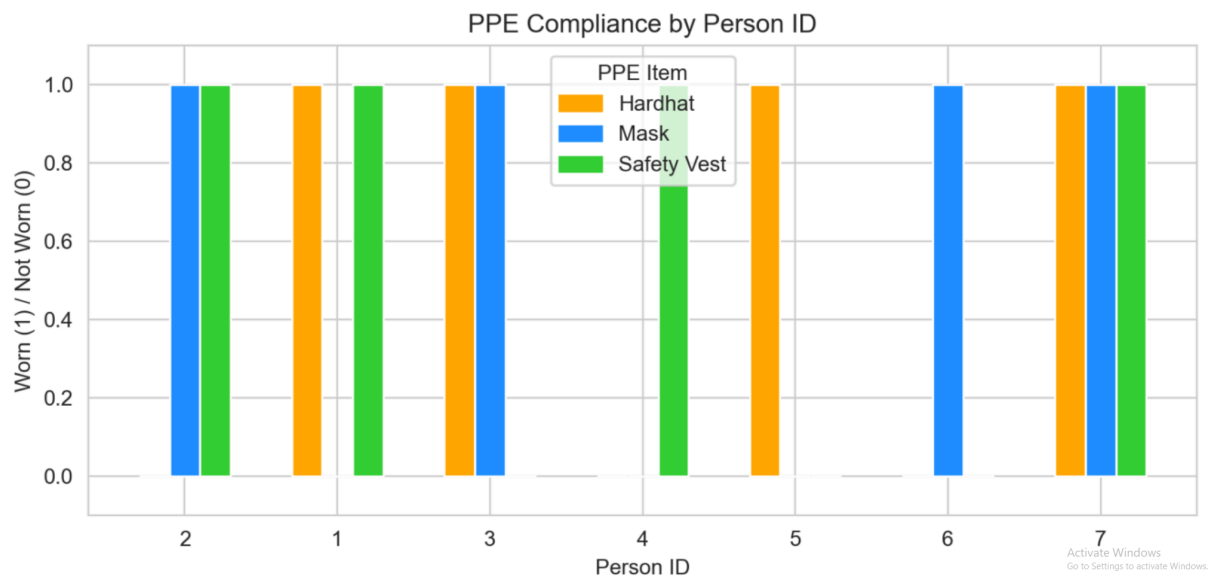
2 Attachments • Scanned by Gmail ⓘ



6.4 Analytics Dashboard

The Analytics tab provides several ways to interpret compliance data:

- **Violations by Person ID Chart:**
 - X-axis: Person IDs of tracked workers
 - Y-axis: Number of violations
 - Color coding: Different types of violations
 - Interpretation: Taller bars indicate frequent violations; multiple colors indicate multiple violation types
- **Safety Check Summary Chart:**
 - X-axis: Pass/Fail categories
 - Y-axis: Count of occurrences
 - Interpretation: Relative size of bars shows overall compliance rate
- **Raw Data Table:**
 - Provides detailed view of all recorded events
 - Sortable by any column
 - Searchable for specific values
 - Exportable for external reporting



7. Maintenance

7.1 Database Management

The system uses CSV files for data storage:

- **Data Location:**
 - webcam_ppe_tracking.csv
 - video_ppe_tracking.csv
 - Both located in the root directory of the application
- **File Format:**
 - Comma-separated values (CSV)
 - Columns: Timestamp, Person ID, Equipment Worn, Equipment Not Worn
- **Data Management:**
 - Backup these files regularly to prevent data loss
 - Archive old files periodically to manage file size
 - Use spreadsheet software for advanced data manipulation

7.2 Log Files

The system generates several types of log files:

- **Application Logs:** Output in the terminal/command prompt
- **Streamlit Logs:** Located in `~/.streamlit/` directory
- **Error Logs:** Displayed in the terminal and web interface

To enable more detailed logging:

1. Create a file named `.streamlit/config.toml` in your home directory
2. Add the following content:

```
[logger]

level = "info"
```
3. Restart the application

7.3 Backup and Restore

To back up your WorkWithHelmtOnly installation and data:

1. **Database Backup:**
 - Copy the CSV files to a secure location
 - Consider implementing scheduled backups using system tools
2. **Configuration Backup:**
 - Save the `.streamlit/` directory if you've made custom configurations
3. **Full System Backup:**
 - Copy the entire project directory to create a complete backup
 - Include the virtual environment if you have custom dependencies

To restore from backup:

1. Reinstall the application following the installation instructions
2. Replace the CSV files with your backed-up versions
3. Restore any custom configuration files

7.4 System Updates

To update the WorkWithHelmtOnly system:

1. Navigate to the project directory
2. Pull the latest changes from the repository:

```
git pull origin main
```
3. Update dependencies:

```
pip install -r requirements.txt --upgrade
```
4. Restart the application

Note: Always back up your data before performing updates.

8. Troubleshooting

8.1 Common Issues

Webcam Not Working

- **Symptoms:** No video displayed, "Could not open webcam" error
- **Solutions:**
 - Verify the webcam is properly connected
 - Check if another application is using the webcam
 - Try a different USB port
 - Update webcam drivers
 - Test the webcam with another application

Slow Performance

- **Symptoms:** Low FPS, laggy interface, delayed detections
- **Solutions:**
 - Reduce the FPS setting in the interface
 - Lower the resolution of the input video
 - Close other resource-intensive applications
 - Use GPU acceleration if available
 - Reduce the number of people in the frame

Inaccurate Detections

- **Symptoms:** False positives, missed detections, incorrect classifications
- **Solutions:**
 - Improve lighting conditions in the monitored area
 - Ensure clear visibility of workers and safety equipment
 - Adjust the camera angle for better viewing
 - Ensure workers are within a reasonable distance from the camera

Email Notifications Not Sent

- **Symptoms:** No email alerts received for violations
- **Solutions:**
 - Verify the recipient email address is correct
 - Check internet connectivity
 - Ensure the sender email has proper permissions
 - Check spam/junk folders in the recipient's mailbox

8.2 Error Messages

"Error: Could not open webcam"

- **Cause:** The system cannot access the webcam
- **Solution:** See "Webcam Not Working" section above

"Error: Failed to send email"

- **Cause:** Email notification could not be sent
- **Solution:** Check internet connection and email settings

"Error: Could not read frame from webcam"

- **Cause:** Webcam connection was lost during monitoring
- **Solution:** Check webcam connection and restart the application

"No data available for analysis"

- **Cause:** No monitoring data has been recorded yet
- **Solution:** Run monitoring sessions to generate data

8.3 Performance Optimization

To optimize system performance:

- **Hardware Optimization:**
 - Use a computer with dedicated GPU
 - Ensure adequate cooling for extended monitoring
 - Close unnecessary background applications
- **Camera Optimization:**
 - Use good lighting in the monitored area
 - Position camera to minimize complex backgrounds
 - Limit the number of people in the frame
- **Application Settings:**
 - Adjust FPS slider to balance performance and accuracy
 - Process shorter video segments for large files
 - Run analysis during off-peak hours for shared resources

8.4 Support Contact

For additional support:

- **GitHub Issues:** Create an issue at <https://github.com/AbdullahAboElHija/FinalProject/issues>
- **Email Support:** Contact the developers at:
 - Abdullah.abo.el.hija@e.braude.ac.il
 - Zenab.Abed.Elghani@e.braude.ac.il
- **Documentation:** Refer to the project documentation on GitHub

9. Appendices

9.1 Glossary

- **YOLO (You Only Look Once):** Object detection algorithm used for identifying workers and safety equipment
- **SORT (Simple Online and Realtime Tracking):** Algorithm used for tracking workers across video frames
- **PPE (Personal Protective Equipment):** Safety gear that workers must wear, such as helmets and safety vests
- **FPS (Frames Per Second):** Rate at which video frames are processed
- **IoU (Intersection over Union):** Metric used to evaluate object detection accuracy
- **Confidence Score:** Probability value indicating the certainty of a detection
- **Bounding Box:** Rectangle drawn around a detected object
- **Tracking ID:** Unique identifier assigned to each tracked worker

9.2 Keyboard Shortcuts

The following keyboard shortcuts are available in the web interface:

- **R:** Refresh the current page
- **Escape:** Cancel current operation
- **Ctrl+Home:** Return to the main dashboard
- **F11:** Toggle fullscreen mode (browser dependent)

9.3 File Formats

CSV File Structure

The system generates CSV files with the following columns:

- **Timestamp:** Date and time of the detection (YYYY-MM-DD HH:MM)
- **Person ID:** Numeric identifier for the tracked person
- **Equipment Worn:** Space-separated list of detected safety equipment
- **Equipment Not Worn:** Space-separated list of missing safety equipment

Example:

```
2025-04-18 14:32:45,1,Helmet Safety Vest,None
2025-04-18 14:32:45,2,Safety Vest,NO-Helmet
2025-04-18 14:32:45,3,None,NO-Helmet NO-Safety Vest
```

Image Files

- **Violation Snapshots:** JPEG format (violation_snapshot.jpg)
- **Profile Images:** JPEG format (default_profile.jpg)

9.4 API Reference

While the primary interface is the web dashboard, advanced users can interact with the system programmatically:

Key Functions:

- **process_frame(frame):** Process a single video frame for detection
- **save_session_data(csv_file):** Save detection data to CSV
- **send_email_with_attachment(receiver_email, csv_file, image_file):** Send notification email
- **plot_violations_by_person(df, title, container):** Generate violation chart

Note: These functions are not part of a public API and may change in future versions.